SUDHARSAN SM

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Executive Summary

AI/ML Engineer with hands-on experience in **Python**, **SQL**, **FastAPI**, **MongoDB**, **Generative AI**, **Machine Learning**, **Deep Learning** and **Agentic AI**. Built multiple AI-driven and full-stack projects focused on **intelligent automation** and **scalable software solutions with AI**. Participated as an **Analog Astronaut**, gaining exposure to space-mission simulations and mission-oriented problem-solving. **NASA Citizen Scientist** and **Youth Astronomy Congress** prize winner, passionate about applying AI to real-world and mission-driven problems.

Technical Skills

- **Programming**: Python, SQL
- **Frontend Engineering**: HTML, CSS, Bootstrap
- **Backend Engineering**: FastAPI, Pydantic
- **ML Engineering**: TensorFlow, Scikit-learn
- **Data & Visualization**: NumPy, Pandas, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Feature Engineering
- **LLM & SLM Engineering**: LLMs, SLMs, RAG, LangChain, LangGraph, Fine-Tuning, Hugging Face
- **Agent Engineering**: CrewAI, Google ADK, MCP, Tool Calling, Agent Orchestration, Multi-Agent Workflows
- **AI Engineering**: Prompt Engineering, Context Engineering, Harness Engineering, Loop Engineering
- **Workflow Automation**: n8n
- **Data Storage**: MongoDB, MySQL
- **Development & Deployment**: Docker, Git, GitHub

Experience

Zaalima Development, *Python Development Intern* **Oct 2025 – Jan 2026**

- Developed **Python-based** solutions for **automation, web scraping and web application** development, focusing on **real-world problem solving**.
- Built an **automation file organizer**, **a product price tracker with alert system** and **a Flask-based To-Do web application** with database integration.
- Gained hands-on experience in **Flask APIs**, **error handling**, **data extraction**, **Git/GitHub** and **technical documentation**.

Projects

BitScrape: Web Crawling & Data Extraction Platform | *Python, FastAPI, PostgreSQL, Docker*

- Developed a scalable **web crawling and scraping data extraction platform** to discover, fetch, extract and process structured data from web pages.
- Designed and Implemented **asynchronous crawling**, **URL management**, **duplicate detection**, **request handling**, **retry mechanisms** and **persistent data** storage for efficient large-scale web data collection.

AI Voice Assistant | *Python, OpenAI API, Speech Recognition*

- Developed an **AI-based voice assistant** capable of understanding natural language queries and generating intelligent responses.
- Integrated **speech-to-text and text-to-speech** functionality for hands-free interaction and automation tasks.

InBits – AI Powered News & Information Platform | *React, TypeScript, Tailwind CSS, PWA, Scikit-learn*

- Developed a modern **news aggregation platform** that collects, organizes and presents trending and latest news from multiple web sources through an intuitive, feed-based interface.
- Implemented responsive **PWA functionality**, **infinite scrolling**, **category-based discovery** and **personalized content interactions** to provide a fast, mobile-friendly news browsing experience.

Education

Mahendra Institute of Technology **Nov 2022 – May 2026**

Bachelor of Engineering in CSE (Artificial Intelligence and Machine Learning) | **CGPA: 8.37/10**

Achievements

- **Prize Winner** - **Youth Astronomy & Space Science Congress** (YASS) for a research paper presentation
- **First Round Selected** - **Hack2Skill Hackathons** at the **Indian International Science Festival**
- **NASA Citizen Scientist** - **International Astronomical Search Collaboration** (IASC)
- **Innovation Ambassador** – **Institution's Innovation Council** (IIC)

Certifications

- **Hack2Skill - Indian International Science Festival** | Certificate ID: 2024H2S01SIF-SH-P100772
- **AI/ML for Geodata Analysis** – **Indian Institute of Remote Sensing** | Certificate ID: 2024233313872
- **Python Development** – **SaiKet System** | Certificate ID: SKS/A2/C52089
- **Python Programming** – **GUVI** | Certificate ID: 10E5913682T8nLGV28